Deterministic Optimization

Linear Optimization Modeling

Handling Nonlinearity

Andy Sun
Assistant Professor
Stewart School of Industrial and Systems Engineering

The Power of Piecewise Linear Functions

Modeling using Linear Programs

Learning Objectives for this Module

- Expand the scope of linear programming modeling from linear functions to nonlinear functions

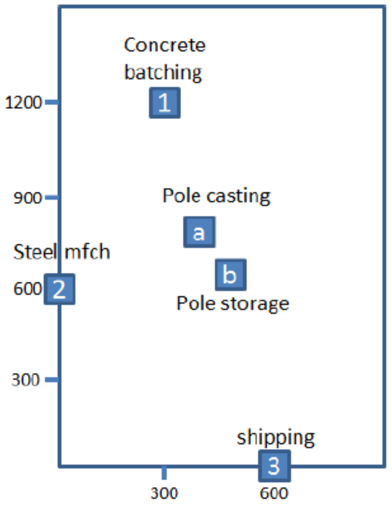
- Discover methods to reformulate convex piecewise linear functions in linear programs

Georgia Concrete Facility Location Problem

- Georgia Concrete (GC) plans to produce a new product concrete poles. It requires two new facilities in its factory, a new concrete casting area (a) to make concrete poles and a storage area (b) for finished product.
- The current layout of GC factory is plotted here:
- Costs of moving parts per unit distance:

1) From 1 to a
2) From 2 to a
3) From 2 to b
4) From a to b
5) From b to 3

Material Handling Cost $/ft	Pole Casting *a*	Pole Storage *b*
Pole casting	-	-
Pole storage	4.00	-
Concrete batch	1.10	-
Steel Mfch	0.70	0.65
Shipping	-	0.40



Metric (Distance) Functions

- Distance between two points is measured as follows:
  - Given $a = (x_1, y_1)$ and $b = (x_2, y_2)$
  - Distance $d_1(a, b) = |x_1 - x_2| + |y_1 - y_2|$
  - Called Manhattan distance or l_1-metric

- Compare with our normal definition of distance:
  - $d_2 = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
  - Called Euclidean distance or l_2-metric

- Manhattan distance makes more sense for GC, as moving large quantity of heavy equipment is done on railroads built in grids.

Nonlinear Optimization Model

- GC wants to minimize the total cost of transporting between facilities
- Decision variables:
    - Locations of new facilities (a) and (b): (x_1, y_1) and (x_2, y_2)
- Facility location model:

- $$\begin{aligned} \min \quad & 4d(x_1, y_1, x_2, y_2) + 1.1d(x_1, y_1, 300, 1200) + 0.7d(x_1, y_1, 0{,}600) \\ & +0.65d(x_2, y_2, 0{,}600) + 0.4d(x_2, y_2, 600{,}0) \\ \text{s.t. } & 0 \le x_i \le 900,\ \forall i = 1{,}2, \\ & 0 \le y_i \le 900,\ \forall i = 1{,}2. \end{aligned}$$

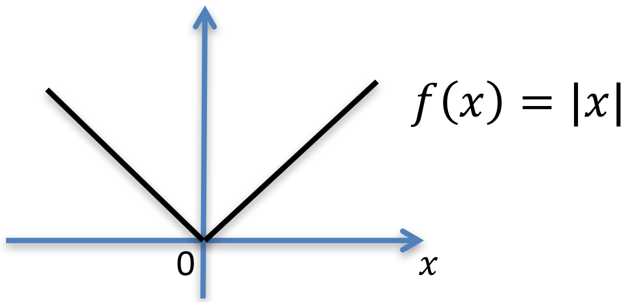
Nonlinear Optimization Model

- $$\min \ 4d(x_1, y_1, x_2, y_2) + 1.1d(x_1, y_1, 300, 1200) + 0.7d(x_1, y_1, 0{,}600)$$
  $$+0.65d(x_2, y_2, 0{,}600) + 0.4d(x_2, y_2, 600{,}0)$$
  s.t. $0 \leq x_i \leq 900,\ \forall i = 1{,}2,$
  $0 \leq y_i \leq 900,\ \forall i = 1{,}2.$

- $$\min \ 4(|x_1 - x_2| + |y_1 - y_2|) + 1.1(|x_1 - 300| + |y_1 - 1200|)$$
  $$+ 0.7(|x_1| + |y_1 - 600|) + \ 0.65(|x_2| + |y_2 - 600|) + +0.4(|x_2 - 600| + |y_2|)$$
  s.t. $0 \leq x_i \leq 900,\ \forall i = 1{,}2,$
  $0 \leq y_i \leq 900,\ \forall i = 1{,}2.$

Absolute Value Function

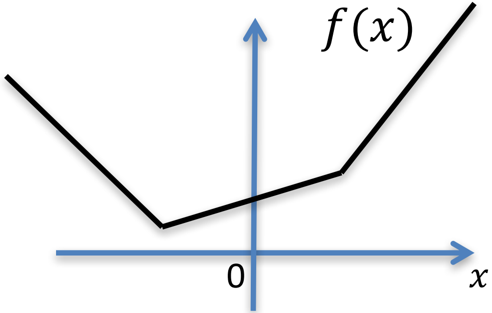
- An absolute value function $f(x) = |x|$ has two linear pieces



- The two linear pieces are x and $-x$.
- $|x|$ can be written as $|x| = \mathbf{max}\{\boldsymbol{x}, -\boldsymbol{x}\}$
- This is called a **piecewise linear function.**
- Clearly, $|x|$ is also a convex function.
- Therefore, $|x|$ is a **convex piecewise linear function**.

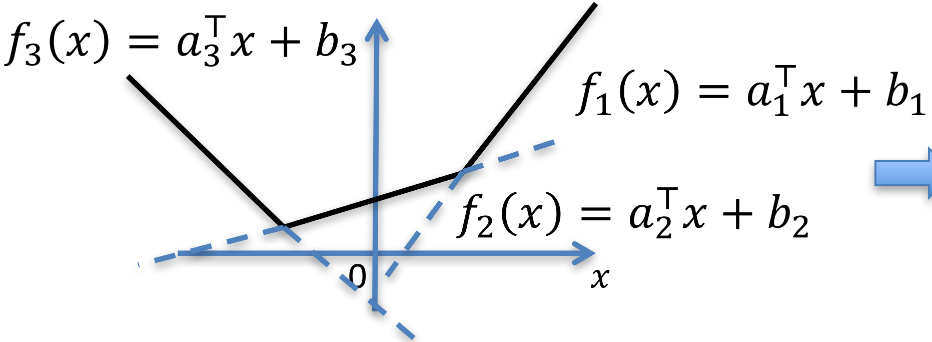
Convex Piecewise Linear Function

- A convex piecewise linear function looks like:



Any convex PWL $f(x)$ can be written as max of a finite number of linear functions

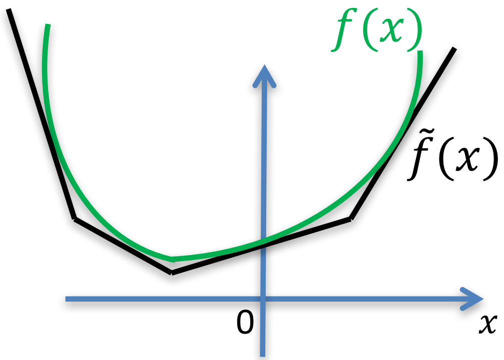
$$f(x) = \max\{a_1^\top x + b_1, \dots, a_m^\top x + b_m\}$$



$$f(x) = \max\big(f_1(x), f_2(x), f_3(x)\big)$$
$$= \max(a_1^\top x + b_1, a_2^\top x + b_2, a_3^\top x + b_3)$$

Convex PWL Approximation

- Any convex function $f(x)$ can be approximated by a convex PWL function $\tilde{f}(x)$ to an arbitrary accuracy.



Summary

- We motivated nonlinear optimization using a facility location problem.

- We introduced l_1 metric function.

- We studied how to represent $|x|$ as a maximum of two linear functions.

- We defined convex PWL and discovered it is a maximum of a finite number of linear function.